Daily Drilling Report
SPD19B-03
9/8/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
43.0	8 1/2"	188.00	72.83	31.70	91.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	264,057.41	815.32	10,611,836.81		38.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,940.00	2,949.00		9.00	5.50	1.6

Operations at Report Time
RIH directional 8 1/2" BHA#11 @ 1100m.

Operations Summary
Directional drilling 8 1/2" hole section f/ 2940m to 2949m. Stop drilling due to low ROP. Pump out of the hole f/ 2949m to 2300m. Normal POOH f/ 2300m to surface. Make 4 stands of 4 3/4" DC & 3 1/2" HWDPs. Rack same in derrick.

Operations next 24hrs
RIH directional BHA#11 to bottom. Resume drilling.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/6/2014	10/22/2014
Trip Drill	9/8/2014	10/29/2014

STOP Cards

Number of Reports: 35

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	06:00	6.00	8 1/2"	D	03	P	Directional drilling 8 1/2" hole section in rotation mode f/ 2940m to 2949m w/ 570 gpm / 2300-2350 psi / 35 kips / 60 rpm / 14-22 Kft-lbf. Decided to POOH due to low ROP. ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Take survey before connection.
06:00	10:00	4.00	8 1/2"	D	06	P	F. Check, well static. B/ Ream f/ 2949m to 2721m (9 5/8" CSG shoe) w/ 450 gpm / 1500 psi / 30 rpm / 12-18 Kft-lbs.
10:00	12:00	2.00	8 1/2"	D	04	P	Pump out of hole f/ 2721m to 2315m w/ 450 gpm/ 1300 psi. F. Check, well static.
12:00	21:00	9.00	8 1/2"	D	04	P	POOH directional drilling BHA#10 on elevator f/ 2315m to surface. R/Back 5 7/8" HWDPs, 6 3/4" NMDC, Float sub. L/D 7" drilling jar. Drain the 6 3/4" mud motor. B/Out the 8 1/2" PDC bit. Rack the motor, PNMDC, 8" ST Stabilizer, 6 3/4" NMDC & Pulser sub in derrick. ** 8 1/2" PDC Bit dull grading: (2-6-RO-N/S-X-1/8-BT-PR) ** 8" ST. stabilizer: In gauge, no balled up. ** Motor bearing gap: 5 mm ** 7" drilling Jar: Good condition ** Sleeve Stab.: 1/16" undergauge
21:00	23:30	2.50	8 1/2"	D	01	P	Change out 5 7/8" pipe handling equipment to 3 1/2" . P/U & RIH 6 jts of 4 3/4" DC & 6 jts of 3 1/2" HWDPs. Flush the string w/ 400 gpm / 80 psi. POOH & Rack same in derrick. Change out to 5 7/8" handling equipment.
23:30	00:00	0.50	8 1/2"	D	04	P	B/out and L/down 6 3/4" old Mud Motor & Pony NMDC. >>> The HPOGC 6 3/4" new mud motor received f/ KISH at 24:00 hrs.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment

00:00	00:00		8 1/2"				Sep 09 , 2014
							
							00:00 - 01:30 P/U, M/U 8 1/2" directional BHA#11 consisting of items below and RIH to 22m. Align Motor with MWD.
							
							1) 8 1/2" New Varel PDC bit (Type: VM713GUX, IADC: M332, SN: 6006868, w/ 4 x 14/32" nozzles)
							2) 6 3/4" Mud Motor (Bent housing: 1.25°, Sleeve Stabilizer: 8 3/8" , Rev./gal: 0.28 , Lobes: 7/8 , Stage: 5)
							3) used 8" COR Stabilizer (SN: COR-2941-AD)
							4) 6 3/4" NMDC
							5) 6 3/4" MWD TMS
							
							01:30 - 02:00 Connect TDS & perform MWD & Motor shallow hole test w/ 500 gpm/ 1100 psi, ok.
							
							02:00 - 04:00 P/U the rest of BHA#11 consisting of items below and RIH to 252m.
							
							06) 6 3/4" NMDC
							07) 6 3/4 F. Sub
							08) XOS (4 1/2" IF pin x XT57 box)
							19) 20 Jts 5 7/8" HWDP
							10) used 7" drilling jar (3021-AD)
							11) 3 Jts 5 7/8" HWDP
							
							>>> Weight below jar (in mud): 38 kips
							>>> Total BHA weight (in mud): 45 kips
							
							04:00 - 06:00 RIH directional drilling BHA#11 on 5 7/8" DP f/ 252m to 1100m.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,949.00	CaCo3/KCL/ Polymer	10.50	60	28.0	22.000	2.7
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.400	1.400	1.300			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	86.0	14.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	11.000	0/100		748.0	2715.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7362.0	30.0	0.0	1.53		12.55214	0.435

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.20	6.	235	8.0	998.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HBJ	Chopper	9/8/2014 10:43	9/8/2014 12:51	PAX IN: 4, PAX OUT:4
Morshed	Supply Boat	9/8/2014 07:00	9/8/2014 07:15	Delivered DD6 Cargo, Dep for Aban 8.
Ahmed 6	Supply Boat	9/7/2014 17:30		Received DD6 Cargo for Storage.Transfrd Aban8 B/L to Moje D.Red 1 Lift cargo f/DD6 for SR2.Stand by.
Moje Darya	Supply Boat	9/8/2014 01:30	9/8/2014 16:35	Delivered Cargo, B/Loaded Cargo for Kish, Dep for Ahan 8.
Borkan	Supply Boat	9/8/2014 23:35	9/9/2014 00:05	Delivered 01 Lift (Mud Motor) f/ Aban-8 & 01 Lift Sahar-2 cargo

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB	5.0	174.0	63.0
Caustic Soda 2014	4.0	500.0	302.0
Soda Ash 25 Kg	4.0	750.0	307.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Drill Water	6.0		314.0
Diesel Fuel	11.0		162.0
Potable Water	23.0	10.0	186.0
Cement G	0.0	0.0	92.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

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Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
10	8 1/2	Varel	VM713GUX	6006867	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	58.17	761,609.40	2,725.00	2,949.00	224.00	3.9	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	575	1,825.0	2,400.0	15	35	160	242
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
51.34	54.77	94.16	5.84	443	365	285	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,940.00	2,949.00	5.50	19,000.0	570	60	2,300.0	35	365

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.89	255.95
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	228.06
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	222.26
X/O	6 1/2	2.813	0.85	35.91
6 3/4" Float sub	6 3/4		0.81	35.06
6 3/4" NMDC	6 3/4	3.250	9.48	34.25
6 3/4" TM - Sub	6 3/4		1.74	24.77
6 3/4" NMDC	6 3/4	3.250	9.48	23.03
8" St. Stabilizer (COR-2941AD)	8	3.000	1.84	13.55
Pony NMDC	6 13/16	3.500	3.09	11.71
6 3/4" Motor (AKO: 1.15° , Sleeve: 8 3/8")	6 3/4		8.34	8.62
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,903.03	251.40	26.20	2,742.69	-364.61	-815.75	893.53	0.64	0.63
2,928.60	251.50	26.30	2,765.62	-368.21	-826.47	904.79	0.13	0.12

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	2,559.14
K1		2,732.00	2,586.65
K2		2,851.80	2,696.55
K3		2,901.00	2,740.86

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Moje Darya to Kish (the B/Loaded items are transferred f/ Ahmed 6 to Moje Darya): (PDF 10 x BB of Sodium Chloride) + (Garbage) + (40 sxs Mica Coarse + 40 sxs Mina Medium + 40 sxs Mica Fine + 30 sxs Nut plug Fine + 30 sxs Nut plug M + 64 sxs Q seal C + 96 sxs Q seal M)

>>> Received:

** Moje Darya f/ Kish: (PDK Wireline tools & Radio Active Source) + (Alghaith 7" Liner & TBG Handling tools) + (Mehran Batch Mixer & accessories) + (MI 54 BBAGs of CaCl2 94-98%) + (135 jts of 7" 23 PPF TBG)

** Borkan f/ ABAN8: (HPOGC 6 3/4" mud motor)

** Morshed f/ Kish: (HPOGC 2 x 4 3/4" Drilling Motor + 2 x float valve + 2 x float sub + 1 x lift sub) + (1 x COR 4 3/4" HM drilling jar)